MICRO GOAL: SECURE GOAL MANAGEMENT BACKEND API

TEAM ID:SWT ID-2026-5026

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Introduction

Goal management plays an important role in improving productivity and organization. Many people struggle to complete large tasks because they are not broken down into smaller achievable goals. The Micro Goal system addresses this problem by allowing users to create and track small tasks that gradually lead to larger achievements.

The project focuses on developing a backend API that securely manages goal-related data. Users can add goals, update their status, and remove completed goals. The system is designed with a simple frontend interface that communicates with the backend server through HTTP requests.

Objectives

- To design a secure backend API for managing user goals.
- To allow users to create and track micro goals.
- To provide CRUD operations (Create, Read, Update, Delete) for goal management.
- To store goal information in a structured database.
- To demonstrate API-based web application development.

System Architecture

The system architecture consists of three major components:

1. Frontend Interface The frontend provides the user interface where users can add and view goals. It is developed using HTML, CSS , and JavaScript.
2. Backend API Server The backend is responsible for processing requests from the frontend. It manages business logic and communicates with the database. The backend is developed using Python and the Flask framework.
3. Database The database stores user information and goal data. SQLite or MySQL can be used to maintain records of goals, descriptions, and status.

Technologies UsedFrontend Technologies:

- **HTML**
- **CSS**
- **JavaScript**

Backend Technologies:

- **Python**
- **Flask Framework**

Database:

- **SQLite / MySQL**

Tools:

- **Postman for API testing**
- **GitHub for project version control**

Database DesignThe system database consists of two main tables:

Users Table:

Stores user registration information such as username, email, and password.

Goals Table:

Stores goal details including goal title, description, status, and creation date.

This structure allows each user to manage multiple goals efficiently.

API Endpoints

POST /addgoal Used to create a new goal.

GET /goals Retrieves all goals stored in the database.

PUT /update/<id> Updates the status of an existing goal.

DELETE /delete/<id> Deletes a goal from the system.

Advantages

- Helps users manage tasks effectively.
- Encourages productivity through small achievable goals.
- Easy to use interface.
- Secure backend data management.
- Scalable architecture that can be extended with more features.

Future Enhancements

- User authentication and login system.
- Mobile application integration.
- Goal reminders and notifications.
- Progress analytics dashboard.
- Cloud deployment for global access.

Conclusion

The Micro Goal Secure Goal Management Backend API provides an efficient way to organize tasks and track progress.

By combining a simple user interface with a powerful backend API, the system demonstrates the advantages of modern web development practices.

The project successfully implements goal creation, tracking, updating, and deletion functionalities. It can be further extended with authentication, analytics, and cloud deployment features to create a full-scale productivity management system.